

Revenue Command Center

Integration Plan for Global ERP

Date: March 30, 2026 **Module:** Revenue Command Center (RCC) **Status:** Proposed

1. Context

The **Revenue Command Center** is a 5-tab analytics dashboard designed for end-to-end lead-to-cash performance tracking. It provides real-time visibility into revenue generation, lead source ROI, pipeline health, revenue leakage, and AI-powered intelligence signals.

The goal is to build this as a new module in Global ERP, leveraging existing data infrastructure including leads, invoices, PIS advisor scores, and the AI Intelligence Layer (8 engines).

2. Dashboard Overview

The Revenue Command Center consists of 5 tabs:

Tab	Purpose	Key Widgets
Command Center	Executive overview	5 KPI cards, Lead Sources donut chart, Live Pipeline funnel, Agent Revenue Leaderboard, Revenue Heatmap, AI Revenue Intelligence signals
Lead Sources	Source ROI analysis	End-to-end performance table per source, Source summary cards with KPIs
Pipeline	Funnel health	8-stage lead lifecycle, Stage cards with drop-off %, Source-to-Stage conversion matrix
Leakage & Fix	Revenue recovery	Leakage by stage with causes and AED lost, Prioritized Fix Roadmap
AI Engine	Intelligence hub	All AI signals ranked by impact, 7 engine status indicators

2.1 Tab 1 - Command Center (Overview)

Top KPI Cards:

- Total Revenue (AED)
- Revenue Per Lead (RPL)
- Lead-to-Sale Rate (%)
- Cost Per Acquisition (CPA)
- Revenue Leakage (AED)

Widgets:

- **Lead Sources Donut Chart** - Distribution across 8 sources (WhatsApp, Mobile App, Inbound Calls, Outbound Calls, Walk-In, Website, Referral, Social Media)
- **Live Pipeline** - Horizontal funnel: New Lead -> Contacted -> Qualified -> Quote Sent -> Booked -> Showed Up -> Invoiced -> Cash Collected, with drop-off % and AED values at each stage
- **Agent Revenue Leaderboard** - Ranked table with leads count, conversion %, RPL; toggleable by Revenue / RPL / Conv%
- **Revenue Heatmap** - Revenue by hour x day-of-week grid (7 days x 12 hours)
- **AI Revenue Intelligence** - Top signals with severity (CRITICAL/HIGH/MED), category tags, impact AED, confidence %

2.2 Tab 2 - Lead Sources

- **Lead Source Performance Table** - Columns: Source, Leads, Qualified, Booked, Showed, Converted, Revenue, RPL, CPA, ROI. Toggleable views: Revenue, RPL, ROI, Conv%
- **Source Summary Cards** - Per-source cards showing Leads, Converted %, Revenue, RPL, CPA, ROI with mini progress bars

2.3 Tab 3 - Pipeline

- **Lead Lifecycle Pipeline** - Full 8-stage funnel visualization with stage counts and drop-off percentages
- **Stage Cards** - 8 cards (New Lead through Cash Collected) showing count, AED value, drop %, leads lost
- **Source-to-Stage Conversion Matrix** - Table showing conversion rate at each stage transition broken down per source

2.4 Tab 4 - Leakage & Fix

- **KPI Cards:** Total Leakage AED, Leads Lost count, Biggest Leak stage, Fastest Fix recommendation
- **Revenue Leakage by Stage** - Each stage transition showing: cause, leads lost, AED lost, suggested fix
- **Fix Roadmap** - Prioritized recovery plan with effort level (Low/Medium/High), timeline, and recoverable AED

2.5 Tab 5 - AI Engine

- **KPI Cards:** Total AI Impact AED, Active Signals count, Avg Confidence %, Engines Active (7/7)
- **AI Revenue Intelligence Signals** - Full list sorted by recoverable revenue impact
- **AI Engine Status** - 7 engines with active/inactive status: Revenue Leak Detection, Source ROI Optimizer, Channel Shift Engine, Booking Drop Analyzer, Upsell Intelligence, Referral Growth Engine, Anomaly Detection

3. Data Mapping to Existing ERP

3.1 Pipeline Stages (8-stage funnel mapped to existing tables)

Dashboard Stage	Query Source	Logic
New Lead	leads	COUNT(*) WHERE created_at BETWEEN from AND to
Contacted	leads + call_sessions	Leads with completed call session
Qualified	leads	lead_status IN ('accepted', 'processing', 'car_in', 'closed_won')
Quote Sent	estimates	Estimates linked to leads via lead_id
Booked	lead_bookings	Active or completed bookings
Showed Up	leads	checkin_at IS NOT NULL (vehicle checked in)
Invoiced	invoices	Invoices linked to leads via lead_id
Cash Collected	invoices	paid_at IS NOT NULL

Reference Template: `packages/ai-core/src/pis/funnelRepository.ts` (existing 9-stage funnel uses identical query patterns)

3.2 Lead Source Mapping

Current LeadSource type (`packages/ai-core/src/crm/leads/types.ts`): `call` | `website` | `ads` | `whatsapp` | `walk_in` | `referral` | `other`

Current Value	Dashboard Label	Notes
<code>whatsapp</code>	WhatsApp	Direct mapping
<code>call</code> + <code>direction=inbound</code>	Inbound Calls	JOIN with <code>call_sessions.direction</code>
<code>call</code> + <code>direction=outbound</code>	Outbound Calls	JOIN with <code>call_sessions.direction</code>
<code>walk_in</code>	Walk-In	Direct mapping
<code>website</code>	Website	Direct mapping
<code>referral</code>	Referral	Direct mapping
<code>ads</code>	Social Media	Rename for display
NEW <code>mobile_app</code>	Mobile App	New source value to add

3.3 Agent Leaderboard

Reuse existing `pis_advisor_scores` table which already contains:

- `composite_score` , `conversion_rate` , `revenue` , `tier` (ELITE/STANDARD), `rank`

Supplement with RPL (Revenue Per Lead) computed from `invoices` joined to `leads.agent_employee_id` .

3.4 AI Engine Mapping (7 virtual engines from existing 8 engines)

Dashboard Engine	Maps To	Signal Source
Revenue Leak Detection	E1 (Funnel) + E3 (Revenue)	Leakage signals
Source ROI Optimizer	E3 (Revenue)	Source performance signals
Channel Shift Engine	E3 (Revenue)	Channel comparison signals
Booking Drop Analyzer	E1 (Funnel)	Booking drop-off signals

Upsell Intelligence	E3 (Revenue)	Upsell opportunity signals
Referral Growth Engine	E3 (Revenue)	Referral performance signals
Anomaly Detection	E5 (Anomaly)	Anomaly signals

Orchestrator: `packages/ai-core/src/intelligence/orchestratorService.ts`

4. Implementation Plan

Phase 1: Backend Foundation

4.1 Database Migrations

Migration 184: `rcc_marketing_spend` table

```
Table: rcc_marketing_spend
- id (UUID, PK)
- company_id (UUID, FK -> companies)
- source (TEXT) - matches lead source values
- period_start (DATE)
- period_end (DATE)
- spend_amount (NUMERIC 14,2)
- currency (TEXT, default 'AED')
- notes (TEXT, nullable)
- created_by_user_id (UUID, FK -> users)
- created_at, updated_at (TIMESTAMPTZ)
```

Required for CPA and ROI calculations. Starts empty; populated via manual entry or CSV import.

Migration 185: Permissions

- `rcc.dashboard.view` - View the Revenue Command Center
- `rcc.marketing_spend.manage` - Manage marketing spend data

4.2 New Backend Module

Location: `packages/ai-core/src/revenue-command-center/`

File	Purpose
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<code>types.ts</code>	TypeScript interfaces: <code>RccFilter</code> , <code>RccOverviewData</code> , <code>RccLeadSourceData</code> , <code>RccPipelineData</code> , <code>RccLeakageData</code> , <code>RccAiEngineData</code> , <code>PipelineStage</code> , <code>AgentLeaderboardEntry</code> , <code>LeadSourcePerformance</code> , <code>LeakageStage</code>
<code>repository.ts</code>	SQL query functions (modeled after <code>funnelRepository.ts</code>)
<code>service.ts</code>	Orchestration layer with <code>Promise.all</code> parallelism
<code>aiSignalGenerator.ts</code>	Transforms existing engine signals into RCC format

Repository Functions:

Function	Description	Data Sources
<code>getOverviewKpis(filter)</code>	5 KPI card values	<code>invoices</code> , <code>leads</code> , <code>rcc_marketing_spend</code>
<code>getPipelineFunnel(filter)</code>	8-stage counts + AED values	<code>leads</code> , <code>call_sessions</code> , <code>estimates</code> , <code>lead_bookings</code> , <code>invoices</code>
<code>getLeadSourceDistribution(filter)</code>	Source counts with inbound/outbound split	<code>leads</code> , <code>call_sessions</code>
<code>getAgentLeaderboard(filter)</code>	Agents ranked by revenue	<code>pis_advisor_scores</code> , <code>leads</code> , <code>invoices</code>
<code>getRevenueHeatmap(filter)</code>	Revenue by day x hour	<code>invoices</code>
<code>getLeadSourcePerformance(filter)</code>	End-to-end table per source	<code>leads</code> , <code>estimates</code> , <code>lead_bookings</code> , <code>invoices</code> , <code>rcc_marketing_spend</code>
<code>getConversionMatrix(filter)</code>	Source x Stage rates	<code>leads</code> , all pipeline tables
<code>getLeakageByStage(filter)</code>	Drop-offs x avg deal value	Pipeline tables, <code>invoices</code>
<code>getMarketingSpend(filter)</code>	CRUD operations	<code>rcc_marketing_spend</code>

4.3 Type Updates

Update `LeadSource` in `packages/ai-core/src/crm/leads/types.ts` :

- Add: `mobile_app` , `social_media` , `inbound_call` , `outbound_call`

4.4 Module Export

Add to `packages/ai-core/src/index.ts` :

```
export * as RevenueCommandCenter from './revenue-command-center/service';
```

Phase 2: API Routes

Base path: `apps/web/app/api/company/[companyId]/revenue-command-center/`

Route File	Method	Returns
<code>overview/route.ts</code>	GET	Tab 1: KPIs, sources, pipeline, leaderboard, heatmap, AI signals
<code>lead-sources/route.ts</code>	GET	Tab 2: Source performance table and cards
<code>pipeline/route.ts</code>	GET	Tab 3: Lifecycle, stage cards, conversion matrix
<code>leakage/route.ts</code>	GET	Tab 4: Leakage KPIs, stage leakage, fix roadmap
<code>ai-engine/route.ts</code>	GET	Tab 5: AI KPIs, signal list, engine status
<code>marketing-spend/route.ts</code>	GET, POST	Marketing spend CRUD

Pattern: Same as `apps/web/app/api/company/[companyId]/pis/master/route.ts` (auth via `getCurrentUserIdFromRequest` , date parsing, service call, JSON response).

Phase 3: Frontend

Page Location: `apps/web/app/company/[companyId]/revenue-command-center/page.tsx`

Architecture:

- Single `"use client"` component with 5 client-side tabs
- Lazy data fetching per active tab

- Dark theme using existing Tailwind classes
- Recharts library for chart visualizations

Component Breakdown:

Tab	Components
Command Center	KPI row (grid-cols-5), PieChart (donut), Pipeline bars, Leaderboard table, Heatmap grid, Signal cards
Lead Sources	Performance table (sortable), Source cards (with progress bars)
Pipeline	Funnel visualization, 8 stage cards, Conversion matrix table
Leakage & Fix	4 KPI cards, Leakage stage list, Fix roadmap list
AI Engine	4 KPI cards, Signal list (expandable), 7 engine status cards

Phase 4: Navigation Integration

Add entry to `packages/ui/src/layout/sidebarConfig.ts` in the company Main section:

```
{
  label: "Revenue Command Center",
  href: "/company/[companyId]/revenue-command-center",
  permissionKeys: ["rcc.dashboard.view"]
}
```

5. Files to Create / Modify

New Files (13)

File Path
<code>packages/ai-core/migrations/184_rcc_marketing_spend.sql</code>
<code>packages/ai-core/migrations/185_rcc_permissions.sql</code>
<code>packages/ai-core/src/revenue-command-center/types.ts</code>

packages/ai-core/src/revenue-command-center/repository.ts
packages/ai-core/src/revenue-command-center/service.ts
packages/ai-core/src/revenue-command-center/aiSignalGenerator.ts
apps/web/app/api/company/[companyId]/revenue-command-center/overview/route.ts
apps/web/app/api/company/[companyId]/revenue-command-center/lead-sources/route.ts
apps/web/app/api/company/[companyId]/revenue-command-center/pipeline/route.ts
apps/web/app/api/company/[companyId]/revenue-command-center/leakage/route.ts
apps/web/app/api/company/[companyId]/revenue-command-center/ai-engine/route.ts
apps/web/app/api/company/[companyId]/revenue-command-center/marketing-spend/route.ts
apps/web/app/company/[companyId]/revenue-command-center/page.tsx

Modified Files (3)

File Path	Change
packages/ai-core/src/crm/leads/types.ts	Extend <code>LeadSource</code> type
packages/ai-core/src/index.ts	Export new RCC module
packages/ui/src/layout/sidebarConfig.ts	Add navigation entry

6. Existing Code Reuse

What	Existing File	Reuse For
Funnel queries	packages/ai-core/src/pis/funnelRepository.ts	Pipeline stage counting patterns
Service orchestration	packages/ai-core/src/pis/service.ts	Promise.all parallel data fetching

AI signal generation	<code>packages/ai-core/src/intelligence/orchestratorService.ts</code>	Transform existing engine outputs
Advisor performance	<code>pis_advisor_scores</code> table	Agent leaderboard data
API route pattern	<code>apps/web/app/api/company/[companyId]/pis/master/route.ts</code>	Auth, date parsing, response format

7. Key Formulas

Metric	Formula
Revenue Per Lead (RPL)	<code>SUM(invoices.grand_total) / COUNT(DISTINCT leads.id)</code>
Lead-to-Sale Rate	<code>COUNT(invoices with paid_at) / COUNT(leads) * 100</code>
Cost Per Acquisition (CPA)	<code>SUM(rcc_marketing_spend.spend_amount) / COUNT(leads with closed_won)</code>
ROI per Source	<code>(Revenue - Spend) / Spend * 100</code>
Revenue Leakage	<code>SUM over stages: (leads_lost_at_stage * average_deal_value)</code>
Average Deal Value	<code>SUM(invoices.grand_total) / COUNT(invoices)</code>
Stage Drop-off %	<code>(count_stage_N - count_stage_N+1) / count_stage_N * 100</code>

8. Risks & Mitigations

Risk	Impact	Mitigation
CPA/ROI requires marketing spend data	CPA/ROI show as N/A until populated	Add simple admin form + CSV import; show "No spend data"

		placeholder
Inbound/outbound call split	Leads without call_sessions fall back to generic "call"	Use LEFT JOIN; classify as "Other Calls" if no direction found
Revenue Leakage is estimated	Average deal value may overstate low-value lead leakage	Document methodology in UI tooltip; consider per-source averages
Large data volumes	Slow queries on companies with 10K+ leads	Add date-range indexes; cache results with 5-min TTL

9. Verification Plan

1. Run migrations 184-185, verify tables created in PostgreSQL
2. Seed `rcc_marketing_spend` with sample data for testing
3. Hit each API endpoint with `?from=2026-03-01&to=2026-03-30`, verify JSON responses
4. Load page at `/company/{companyId}/revenue-command-center`
5. Verify all 5 tabs render correctly with real data
6. Cross-reference pipeline counts with existing PIS funnel (`/pis/funnel`)
7. Confirm AI signals generate correctly from existing engines (E1, E3, E5)
8. Test leaderboard sorting toggles (Revenue / RPL / Conv%)
9. Verify permission gating (`rcc.dashboard.view`)

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